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B.Tech. Degree III Semester Examination November 2017

ME 15-1305 METALLURGY AND MATERIAL SCIENCE (2015 Scheme)

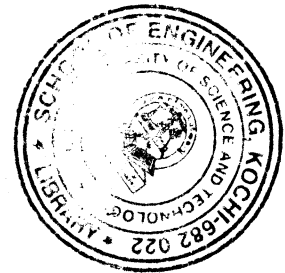
Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Lithium at 20°C is BCC structure and has a lattice constant of 0.35092 nm. Calculate the atomic radius of a lithium atom in nanometers.
- (b) In the solidification of a metal, what is the difference between an embryo and a nucleus? What is the critical radius of a solidifying particle?
- (c) Describe any two surface treatments of steel. Why are these treatments carried out?
- (d) Draw a continuous-cooling transformation diagram for a eutectoid plain-carbon steel.
- (e) What is the difference between the slip and twinning mechanisms of plastic deformation of metals?
- (f) What type of phase diagram is necessary for a binary alloy to be precipitation – hardenable. Comment on the process.
- (g) What are some of the typical characteristics of ceramic materials?
- (h) What are the four basic types of stainless steels?
- (i) Explain the applications of shape memory alloys.
- (j) Compare between recovery and recrystallisation process.



PART B

(4 × 10 = 40)

- II. (a) Explain how the atomic density of crystallographic planes and density can be determined. What is the importance of highly packed planes and directions?
- (b) Distinguish between line and planar defects.

OR

- III. (a) Explain how the equi-axed and columnar grains can be formed during solidification process.
- (b) Explain the Fick's law of diffusion and its importance.
- IV. (a) Reconstruct the iron-carbon phase diagram indicating all key phases, reactions and microstructures.
- (b) Describe various forms of metallic solid solutions and explain their differences.

OR

- V. (a) Distinguish between martempering and austempering process.
- (b) Compare annealing with normalizing process.

(P.T.O.)

- VI. (a) Explain the difference between failure of a material by creep and that by stress rupture.
(b) What are the characteristic microstructural features associated with fatigue fracture?

OR

- VII. (a) Compare and contrast strain hardening with grain boundary strengthening.
(b) Distinguish between elastic and visco elastic behavior.
- VIII. (a) Describe the classification and general properties of cast irons. What are some of their applications?
(b) Why are titanium and its alloys having special engineering importance for aerospace application?

OR

- IX. (a) What are some of the important properties of unalloyed copper that makes it an important industrial metal?
(b) Why are composites finding increased use in air crafts and automobiles?
(c) Comment on the importance of nano materials.

